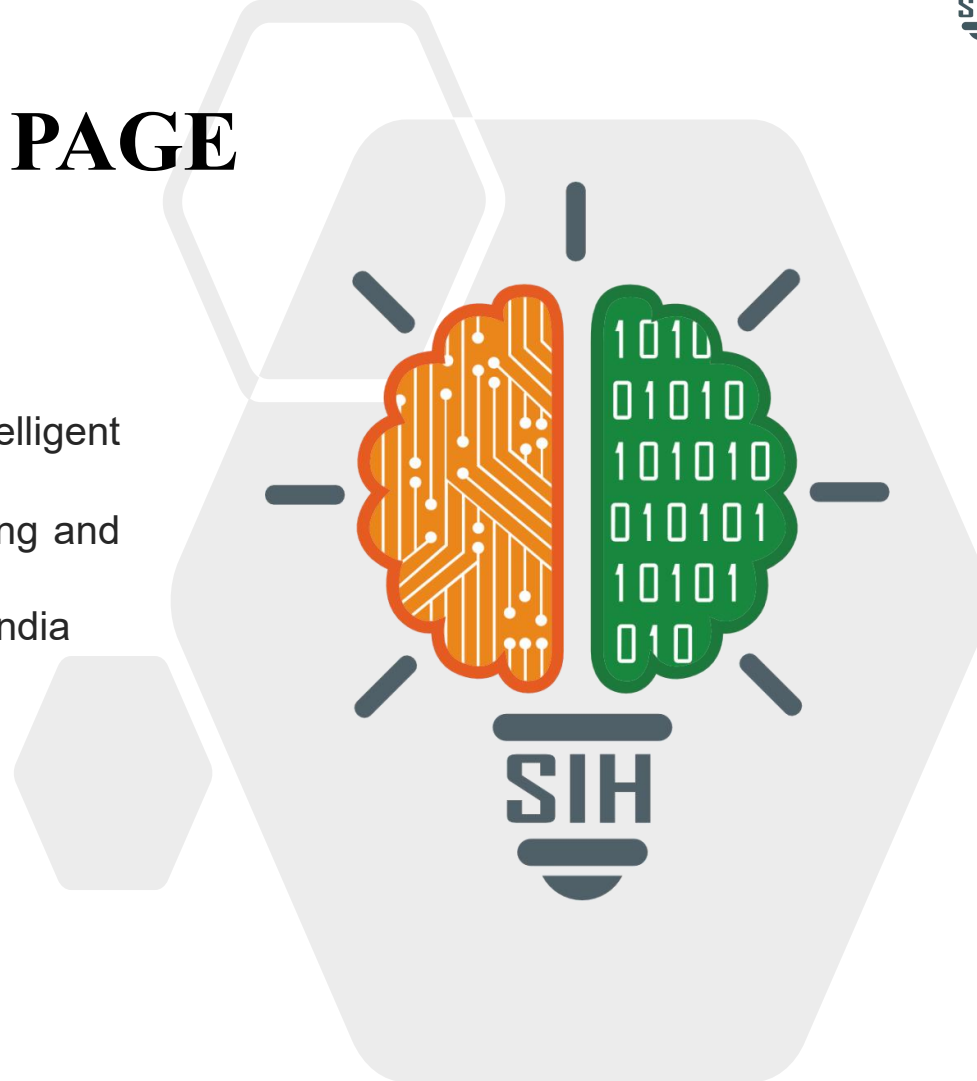


TITLE PAGE

- **Problem Statement ID** – 26006
- **Problem Statement Title** – Development of an Intelligent Freight Forecasting Model for Optimized Vessel Chartering and Bulk Cargo Procurement from overseas to East Coast of India
- **Theme** – Transportation & Logistics
- **PS Category** – Software
- **Team ID** – 153339
- **Team Name (Registered on portal)** – SINISTER SIX



FreightSaarthi

Forecast-driven “Charter Ladder” for SAIL’s coal imports to India’s East Coast

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution

- **Forecasts** Capesize, Panamax, Supramax & Handysize freight 1–26 weeks ahead as calibrated P10–P50–P90 bands
- **Checks** every ship against load-port, discharge-port and strait limits → real parcel size + binding constraint
- **Translates** forecasts into landed \$/t for all 10 origins × 7 East-Coast ports (freight + demurrage + stockyard + lighterage)
- **Optimises** contract mix (COA / period TC / timed spot) and entry timing under risk: $\min E[\text{cost}] + \lambda \cdot \text{CVaR}$
- **Protects** operations: idle-time & ballast planning, early warnings for volatility, congestion, cyclones, Red Sea
- **Delivers** a working web app: decision cockpit, 35-endpoint API, audit ledger, real-data upload & retrain

How it addresses the problem

	Today (daily spot)	With FreightSaarthi
Market entry	Reactive, daily market exploration	Forecast-timed windows, re-solved weekly
Contracts	268 single spot fixtures	54 spot; 73% on multi-voyage COA/TC
Vessel type	Habit (Panamax)	Draft-aware optimum per port pair
Idle time	Unplanned waits, ballast legs	Congestion forecast, laycan spacing, relet / JIT
Risk	Surprises after fixing	Early warnings + CVaR-capped exposure

Working prototype (live cockpit)



Overview page: next-quarter plan for the pilot lane Hay Point → Paradip, backtest KPIs and class forecasts (FastAPI + SQLite + JS cockpit)

Innovation and uniqueness of the solution

1

Physics-informed freight translator

Only 4 class TCEs + bunker are forecast; any lane’s \$/t = A·TCE + B·bunker + C → new routes work from day one.

2

CVaR “Charter Ladder”

Two-stage stochastic LP decides how much COA / time charter to lock each week (±10% MOLCHOPT, dead-freight, ≤35% per tranche).

3

Draft–Tide–Monsoon twin

Load, discharge & strait drafts (Torres 12.2 m, Suez), LOA/beam, berth DWT, stockyard, lighterage, two-port discharge.

4

Honest, decision-focused evidence

Tuned on money saved, not MAE; walk-forward backtest, placebo market shows no fake edge, bootstrap CIs.

5

Broker Quote X-ray + idle manager

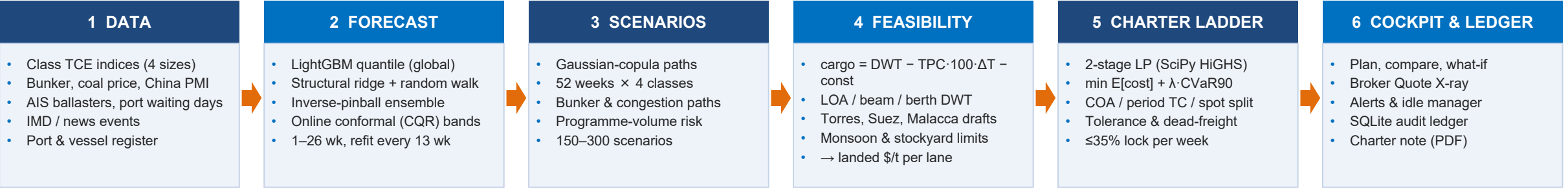
Percentile of any broker quote in the fair-value band; triangulation, relet, JIT slow-steaming, laycan spacing.

6

Audit-ready & real-data-ready

Hash-stamped decision ledger + printable charter notes (CVC/CAG); upload SAIL history → full retrain in ~35 s.

Methodology and process for implementation (Flow Charts/Images/ working prototype)

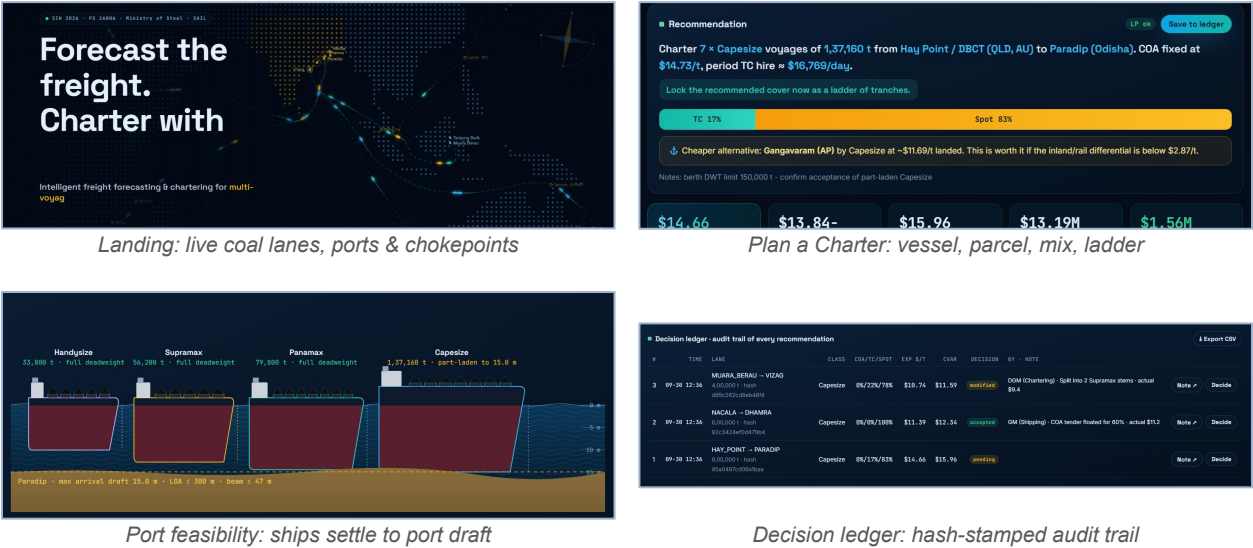


Feedback loop: walk-forward backtest + decision-focused tuning set λ (risk), δ (stopping) and ladder step on a validation period; ledger outcomes (actual fixed rates) recalibrate owner premiums; LP re-solved every week.

Technologies to be used (e.g. programming languages, frameworks, hardware)

Layer	Technology	Used for
Languages	Python 3.14, JavaScript (ES2020), SQL	Models, API, cockpit
Data & ML	pandas, NumPy, LightGBM, scikit-learn	Quantile GBM, ridge, features
Optimisation	SciPy + HiGHS linear programming	CVaR ladder LP in ~30 ms
Backend	FastAPI, Uvicorn, Pydantic, SQLite	35 REST endpoints, audit DB
Frontend	HTML5 Canvas / SVG, Plotly.js	Cockpit, lane map, fit simulator
AI (optional)	Claude API structured output; Chronos / TimesFM	News → events; extra member
Hardware	Any laptop / on-prem server, no GPU	Offline, data stays in SAIL
Quality	Automated API test suite (httpx)	11/11 test groups passing

Working prototype – screens from the running application



Analysis of the feasibility of the idea

Technical

PROVEN

Full prototype built & tested: 35 endpoints, 11/11 test groups, LP solves in ~30 ms, whole pipeline retrains in 35 s–15 min on a laptop.

Data

READY

Every input has a real or public proxy (IMF PortWatch, World Bank, IMD); CSV upload retrains on SAIL's own fixture history.

Economic

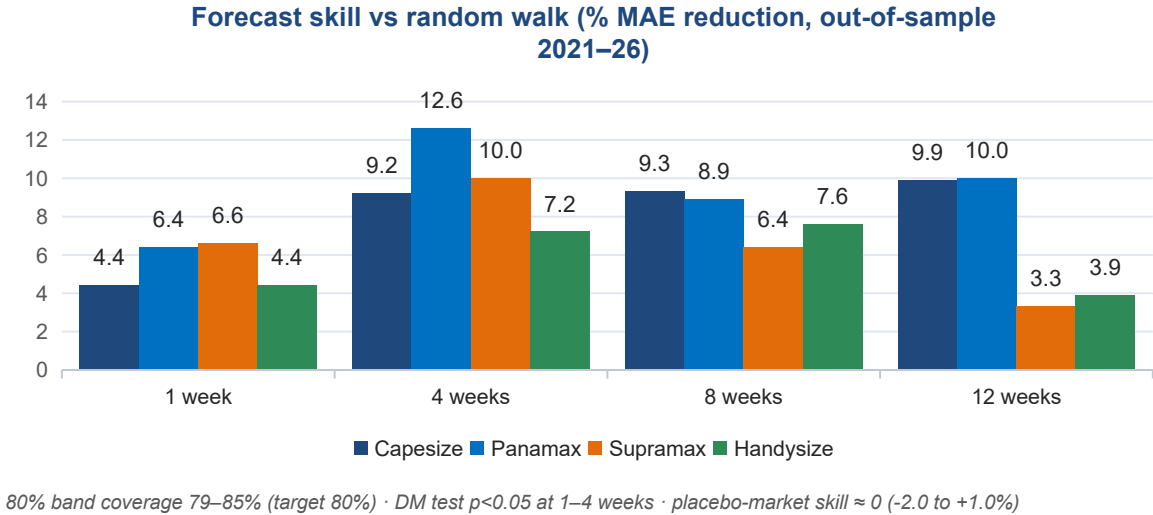
LOW COST

Open-source stack, no licence cost except optional Baltic data. Backtest: \$49.7 M saved on one 900 kt/quarter lane.

Operational

FITS PROCESS

Matches the weekly chartering cadence and tender route: outputs vessel size, tenure, volume and laycan spacing for tenders.



Potential challenges and risks → Strategies for overcoming these challenges

#	Challenge / risk	What could go wrong	Strategy for overcoming it
1	Freight indices are licensed; SAIL history not yet shared	Model trained on proxy data only	Data Hub: validated CSV upload + automatic retrain; public proxies (IMF PortWatch, World Bank); proxy-filled columns labelled
2	Freight is hard to predict beyond 8–12 weeks	Over-confident lock-ins	Calibrated conformal bands, CVaR penalty, ≤35% per ladder tranche, weekly re-solve, placebo & ablation checks
3	Port limits change (dredging, tides, monsoon)	Wrong vessel, part cargo, idle berth	Constraint register with confidence + 'verify_with' source, monsoon draft deltas, congestion & draft alerts
4	COA / TC availability and owner premiums	Quoted price differs from model	Quotes modelled as distributions; ledger records actual fixed rates to learn real premiums
5	PSU procurement & audit rules	Recommendations not actionable	Outputs tender specifications, not bypasses; hash-stamped ledger + printable charter notes for CVC/CAG
6	Desk trust and change management	Tool ignored in practice	Shadow mode first, 'why this recommendation' explanations, what-if levers, one-click scenario comparison

13.9%

lower landed freight vs daily spot
90% CI 11.4–16.3%

\$49.7 M

saved over 22 quarters on one 900 kt/quarter lane
67% of perfect-foresight saving

268 → 54

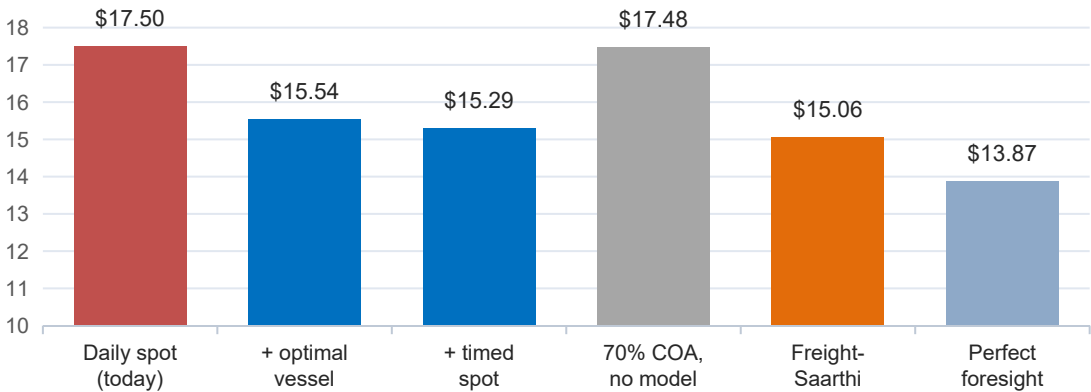
single spot fixtures (~80%)
73% of volume on multi-voyage COA/TC

\$20.6 → \$17.7

bad-quarter freight, \$/t (CVaR90)
cheaper than today in 100% of quarters

Walk-forward backtest, Hay Point (Queensland) → Paradip, test period 2021–mid 2026, synthetic market data: demonstrates the method, not realised SAIL savings.

Average landed freight by strategy (\$/t)



Potential impact on the target audience

Stakeholder	Impact
Chartering desk	From daily market-chasing to a weekly plan; Quote X-ray strengthens every negotiation
Head of Shipping / Procurement	Multi-voyage COA/TC strategy with tender timing, tenure and volume specifications
Finance / CFO	Predictable freight budgets; freight per t of hot metal (\$11.43) and CVaR as KPIs
Steel plants (Bokaro, Rourkela, Bhilai...)	Reliable coal arrivals, fewer stock-outs and demurrage days
Ports (Paradip, Dhamra, Haldia...)	Laycan spacing avoids berth bunching and anchorage queues
Auditors (CVC / CAG)	Every recommendation stored with input hash, data version and officer decision

Benefits of the solution (social, economic, environmental, etc.)

Economic

- Lower and steadier \$/t freight; fewer demurrage and idle days
- Freight-adjusted FOB break-even sharpens origin choice for procurement
- Scales to every origin × East-Coast port with no route-specific model

Social & governance

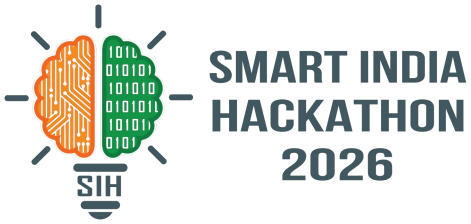
- Transparent, auditable PSU decisions (charter notes, ledger)
- In-house analytics capability (Aatmanirbhar), no vendor lock-in
- Supports National Steel Policy 2017 target of 300 MTPA by 2030-31

Environmental

- JIT slow-steaming ≈ 380 t fuel & ≈ 1,180 t CO2 saved per 6-day Capesize idle window
- Right-sized parcels and fewer ballast legs cut emissions per tonne
- Supports IMO CII (carbon-intensity) compliance of chartered ships



RESEARCH AND REFERENCES



Details / Links of the reference and research work

Methods & research

[1] Rockafellar, R.T. & Uryasev, S. (2000). Optimization of Conditional Value-at-Risk. Journal of Risk 2(3), 21–41. *CVaR ladder*

[2] Romano, Y., Patterson, E. & Candès, E. (2019). Conformalized Quantile Regression. NeurIPS. arxiv.org/abs/1905.03222 *calibrated bands*

[3] Ke, G. et al. (2017). LightGBM: A Highly Efficient Gradient Boosting Decision Tree. NeurIPS. *quantile GBM*

[4] Diebold, F.X. & Mariano, R.S. (1995). Comparing Predictive Accuracy. J. Business & Economic Statistics 13(3). *DM test*

[5] Elmachtoub, A. & Grigas, P. (2022). Smart “Predict, then Optimize”. Management Science 68(1). *decision-focused tuning*

[6] Huangfu, Q. & Hall, J.A.J. (2018). Parallelizing the dual revised simplex method. Math. Prog. Computation 10 (HiGHS). *LP solver*

[7] Ansari, A.F. et al. (2024). Chronos, arxiv.org/abs/2403.07815; Das, A. et al. (2024). TimesFM, arxiv.org/abs/2310.10688 *foundation plug-in*

[8] Stopford, M. (2009). Maritime Economics, 3rd ed. Routledge. *freight economics*

Data & domain sources

[9] Baltic Exchange dry-bulk indices (BCI, BPI, BSI, BHSI): balticexchange.com *freight*

[10] IMF PortWatch, AIS-based port calls & disruptions: portwatch.imf.org *congestion*

[11] World Bank Commodity Price Data (Pink Sheet): worldbank.org/en/research/commodity-markets *coal prices*

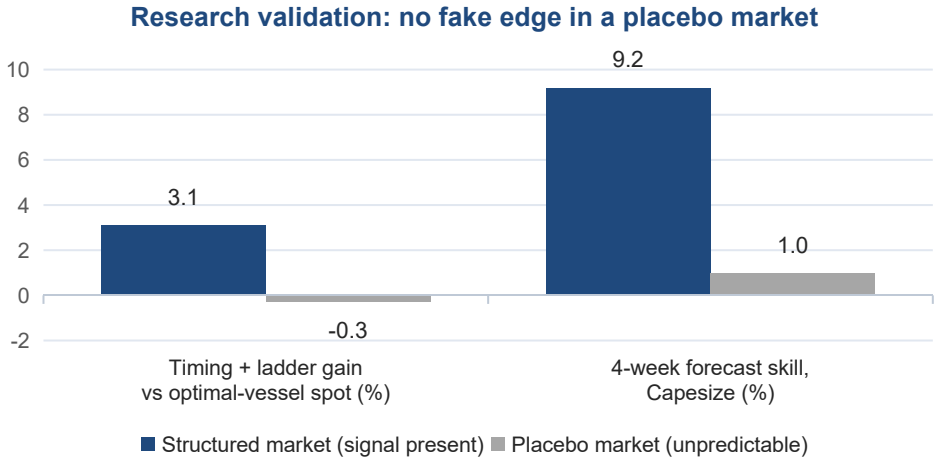
[12] India Meteorological Department, RSMC New Delhi cyclone bulletins: mausam.imd.gov.in *cyclone alerts*

[13] Paradip Port Authority (paradiport.gov.in); SMP Kolkata-Haldia, Dhamra, Gangavaram, Vizag port handbooks *draft/LOA limits*

[14] Ministry of Steel, National Steel Policy 2017: steel.gov.in · [15] IMO Fourth GHG Study 2020 & CII rules: imo.org *policy / emissions*

Our work

[16] FreightSaarthi source code, design document, demo script & API: github.com/Anuttaman216/PS2 *prototype*



Data provenance used in the prototype

Input	Real source	Prototype status
Class TCE indices	Baltic Exchange / brokers	Synthetic → CSV upload
Bunker, coal, PMI	Ship & Bunker, World Bank, NBS	Synthetic, proxy-ready
Port congestion	IMF PortWatch, port reports	Synthetic, proxy-ready
Port / vessel limits	Port handbooks, Equasis	Assumption, flagged
Weather / news	IMD RSMC, news feeds	Rule / LLM extraction

Every assumed number carries a confidence tag and a ‘verify_with’ source in the constraint register; real data replaces synthetic through the Data Hub without code changes.